

# Valence Alignment Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Contrasts on the Exchange-Aligned Data Model

PENPAL Project

2026-05-06

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This notebook compares **valence alignment** across the three experimental conditions using the new exchange-aligned structure.

- **Human-AI**: Human co-writing with an LLM
- **Human-Human**: Two humans co-writing together
- **AI-AI**: Two AI agents co-writing together

Important structural note:

- `author_1` / `author_2` are now treated as **slot 1** / **slot 2** in the cleaned exchange sequence.
- In **Human-AI**, those slots also map onto human vs AI roles.
- In **Human-Human** and **AI-AI**, the slots are positional sides of the exchange, not stable identity labels.

The analysis window is restricted to **complete exchanges** and the default aligned turn range (`analysis_turn` 1 to 9).

**Primary cross-condition object.** Each story contributes one unsigned, label-invariant dyad asymmetry magnitude:

$$A_{\text{valence}}(i) = |\text{atanh}(r_{1 \rightarrow 2, i}) - \text{atanh}(r_{2 \rightarrow 1, i})|$$

where  $r_{1 \rightarrow 2, i}$  is the per-story Pearson correlation between same-turn slot 1 and slot 2 valence, and  $r_{2 \rightarrow 1, i}$  is the correlation between slot 2's previous-turn valence and slot 1's current-turn valence. The main question

is whether  $A_{\text{valence}}$  differs across conditions. The directional-slope LMMs are retained as supplemental views. Starter-mechanism and Human-AI speaker-type decompositions are deferred to a follow-up pass.

**Article-use note.** The primary cross-condition claim should use the unsigned per-story  $A_{\text{valence}}$  test below. Signed directional slot models are diagnostic only: in Human-Human and AI-AI, Slot 1 / Slot 2 are positional labels rather than theoretically meaningful agent identities.

## 1 Load Data

## Data availability:

## # A tibble: 6 x 4

| ## file                                         | `human-ai` | `human-human` | `ai-ai` |
|-------------------------------------------------|------------|---------------|---------|
| ## <chr>                                        | <lgl>      | <lgl>         | <lgl>   |
| ## 1 dyadic_sentiment_scores.parquet            | TRUE       | TRUE          | TRUE    |
| ## 2 interaction_level_surface_metrics.parquet  | TRUE       | TRUE          | TRUE    |
| ## 3 novelty_scores.csv                         | TRUE       | TRUE          | TRUE    |
| ## 4 semantic_exploration_binned.parquet        | TRUE       | TRUE          | TRUE    |
| ## 5 story_embeddings_interaction_level.parquet | TRUE       | TRUE          | TRUE    |
| ## 6 interaction_level_stories_filtered.csv     | TRUE       | TRUE          | TRUE    |

Table 1: Loaded exchange-aligned valence data by condition

| condition   | n_stories | n_turns | min_turn | max_turn |
|-------------|-----------|---------|----------|----------|
| Human-AI    | 100       | 792     | 1        | 8        |
| Human-Human | 36        | 324     | 1        | 9        |
| AI-AI       | 80        | 720     | 1        | 9        |

## 2 Preprocessing

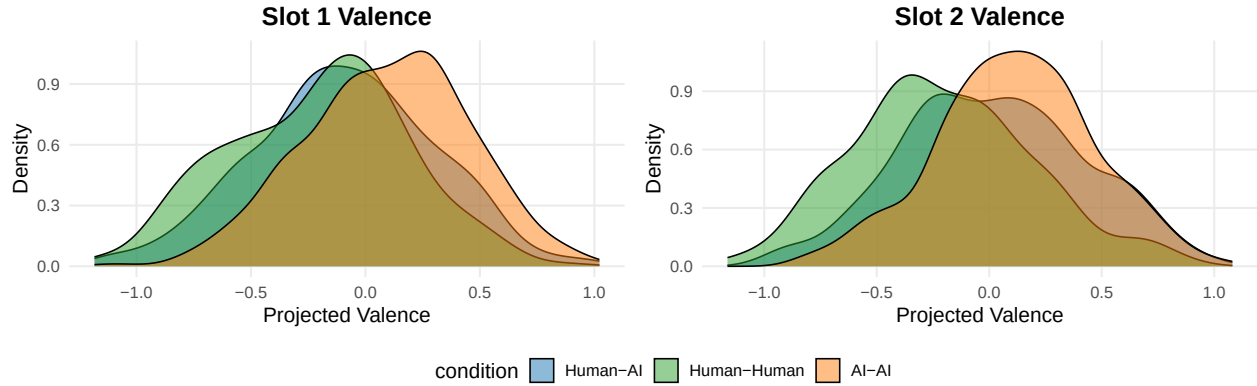
## Analysis rows: 3240

## Stories: 216

## 3 Descriptive Checks

Table 2: Mean projected valence by condition

| condition   | n_stories | n_turns | slot_1_mean | slot_2_mean |
|-------------|-----------|---------|-------------|-------------|
| Human-AI    | 100       | 692     | -0.093      | 0.016       |
| Human-Human | 36        | 288     | -0.208      | -0.207      |
| AI-AI       | 80        | 640     | 0.093       | 0.112       |



## 4 Primary: Per-Story Asymmetry Magnitude

## Stories with both directional correlations available:

Table 3: Per-story A\_valence sample sizes by condition

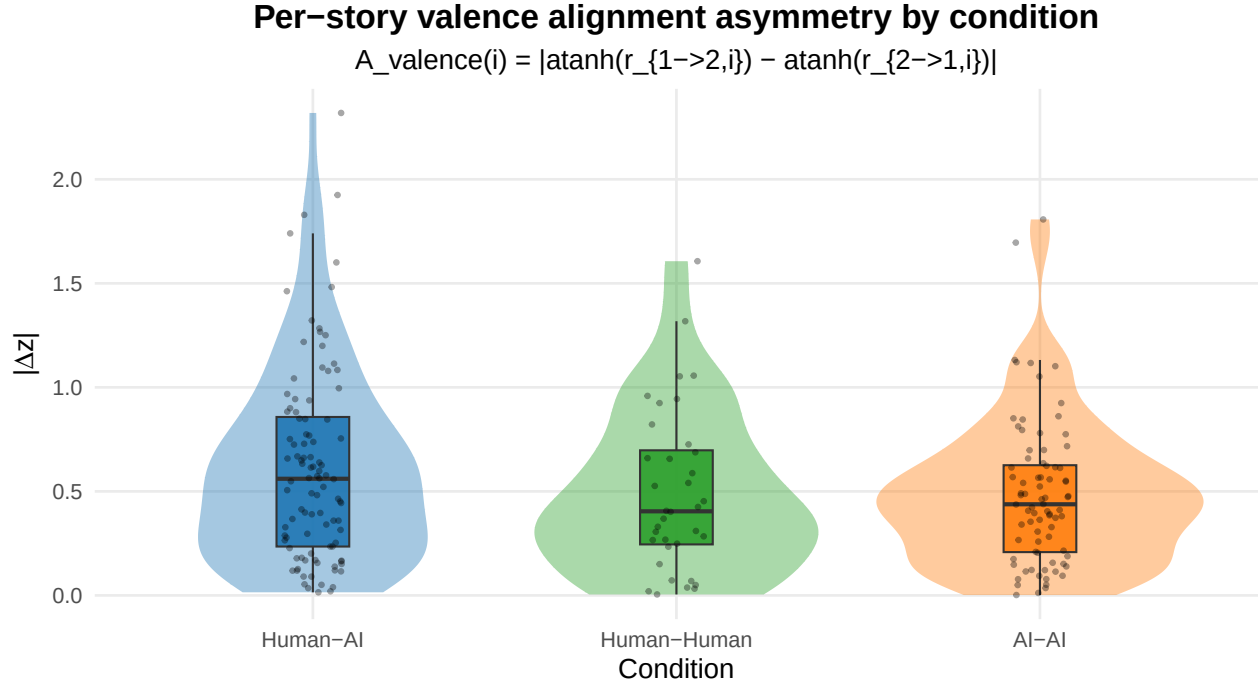
| condition   | n   |
|-------------|-----|
| Human-AI    | 100 |
| Human-Human | 36  |
| AI-AI       | 80  |

Table 4: Per-story |Fisher-z asymmetry| by condition

| condition   | n   | mean_A | sd_A  | median_A |
|-------------|-----|--------|-------|----------|
| Human-AI    | 100 | 0.618  | 0.471 | 0.561    |
| Human-Human | 36  | 0.495  | 0.392 | 0.404    |
| AI-AI       | 80  | 0.485  | 0.353 | 0.438    |

```
## One-way ANOVA: A_valence ~ condition
##           Df Sum Sq Mean Sq F value Pr(>F)
## condition  2   0.91  0.4542   2.602 0.0765 .
## Residuals 213  37.18  0.1745
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Kruskal-Wallis (robust): A_valence ~ condition
##
## Kruskal-Wallis rank sum test
##
## data:  A_valence by condition
## Kruskal-Wallis chi-squared = 3.6397, df = 2, p-value = 0.162
##
## Pairwise Wilcoxon (BH-adjusted):
##
## Pairwise comparisons using Wilcoxon rank sum test with continuity correction
```

```
##
## data: df_asym$A_valence and df_asym$condition
##
##           Human-AI Human-Human
## Human-Human 0.29      -
## AI-AI       0.24      0.85
##
## P value adjustment method: BH
```



## 5 Bootstrap CIs and Effect Sizes on $A\_valence$

Table 5: Bootstrap 95% CI on per-condition mean of  $A\_valence$   
( $R = 10,000$ )

| condition   | n   | mean  | ci_lower | ci_upper |
|-------------|-----|-------|----------|----------|
| Human-AI    | 100 | 0.618 | 0.529    | 0.712    |
| Human-Human | 36  | 0.495 | 0.369    | 0.623    |
| AI-AI       | 80  | 0.485 | 0.413    | 0.564    |

Table 6: Pairwise Cliff's delta on  $A\_valence$

| contrast                | n_a | n_b | cliff_delta | magnitude  |
|-------------------------|-----|-----|-------------|------------|
| Human-AI vs Human-Human | 100 | 36  | 0.147       | negligible |
| Human-AI vs AI-AI       | 100 | 80  | 0.152       | small      |
| Human-Human vs AI-AI    | 36  | 80  | -0.022      | negligible |

## 6 Planned Contrast: $HA$ vs $(HH + AA)/2$

Table 7: Planned contrast: A\_valence in HA vs the average of HH and AA

| contrast                            | estimate | se    | t     | df      | p     | ci_lower | ci_upper |
|-------------------------------------|----------|-------|-------|---------|-------|----------|----------|
| Human-AI vs (Human-Human + AI-AI)/2 | 0.128    | 0.061 | 2.113 | 160.366 | 0.036 | 0.008    | 0.248    |

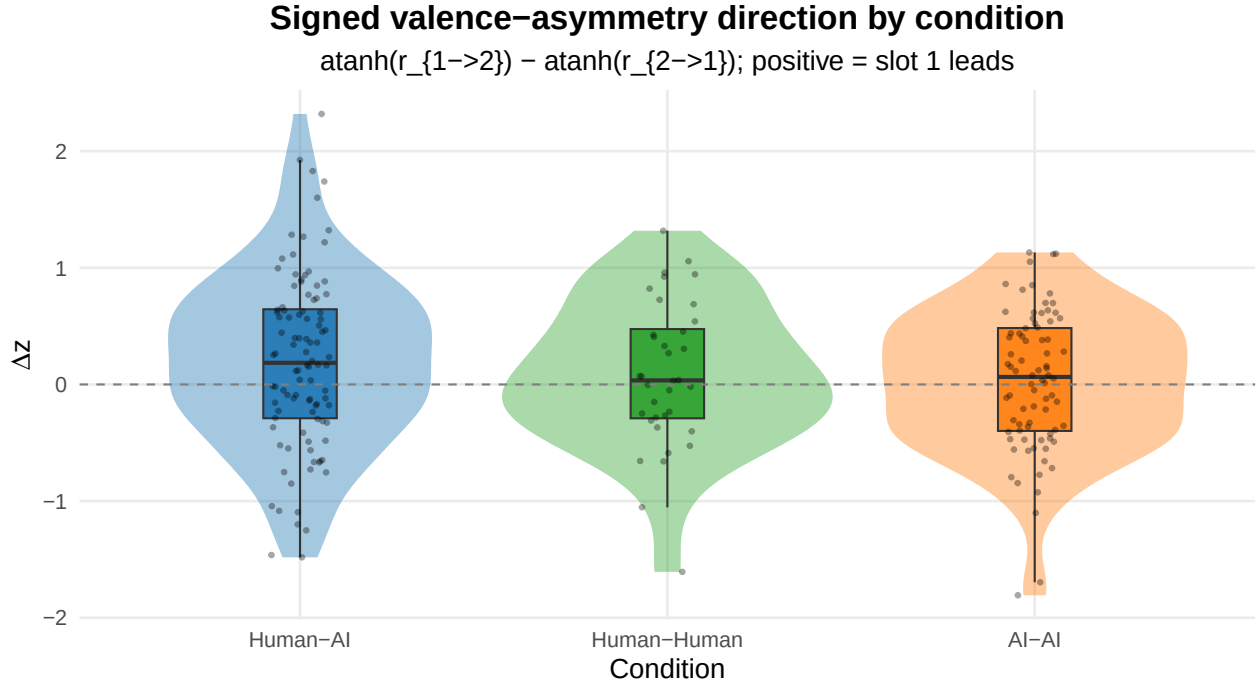
## 7 Directional Stability of Signed Valence Asymmetry

Tests whether the *direction* of the slot-1 vs slot-2 alignment difference is consistent across stories within each condition. The signed quantity is  $\text{signed}_i = \text{atanh}(r_{1 \rightarrow 2, i}) - \text{atanh}(r_{2 \rightarrow 1, i})$ . HA-specific identity-tied directionality predicts high **stability** and **sign\_consistency** in HA, low in HH and AA, even when unsigned A\_valence is similar.

Table 8: Per-condition directional stability of signed valence asymmetry

| condition   | n   | mean  | sd    | stability | sign_consistency | one_sample_t_p |
|-------------|-----|-------|-------|-----------|------------------|----------------|
| Human-AI    | 100 | 0.199 | 0.753 | 0.265     | 0.580            | 0.009          |
| Human-Human | 36  | 0.082 | 0.631 | 0.130     | 0.528            | 0.441          |
| AI-AI       | 80  | 0.024 | 0.602 | 0.039     | 0.550            | 0.727          |

## Brown-Forsythe variance-equality p = 0.1314



## 8 Permutation Test on A\_valence ~ condition

Table 9: Permutation test on  $A_{\text{valence}} \sim \text{condition}$  ( $R = 10,000$  story-label permutations)

| F_obs  | p_perm | R     |
|--------|--------|-------|
| 2.6021 | 0.0755 | 10000 |

## 9 Summary

Primary cross-condition result: per-story  $A_{\text{valence}} = |\text{atanh}(r_{1 \rightarrow 2, i}) - \text{atanh}(r_{2 \rightarrow 1, i})|$ , tested with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon.

Robustness layers added:

- bootstrap 95% CIs on per-condition mean of  $A_{\text{valence}}$
- pairwise Cliff's  $\delta$  effect sizes
- planned single-df contrast HA vs (HH+AA)/2
- 10k story-label permutation test on  $A_{\text{valence}} \sim \text{condition}$

The Human-AI signed Human/AI directional decomposition lives in `analysis/human-ai/valence_alignment_analysis.Rmd` and is not duplicated here.

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized asymmetry across all conditions)

*Analysis completed: 2026-05-06 16:10:21.822097*